Locally spherical regular hypertopes: marked quotients, affine families and a corrected finite table

Human Compression project

Abstract

We give a complete attributed account of the fixed revised manuscript *An exploration of locally spherical regular hypertopes* by Fernandes, Leemans and Ivić Weiss. The spherical classification follows from Maxwell's retained normal-subgroup theorem and explicit antipodal incidence arguments. A reflection-commutator proof gives the affine translation-kernel conclusion under the universal-residue hypothesis, and the displayed toroidal families are joined to their exact parameter domains. A finite separating-set argument supplies arbitrarily large finite examples for every retained compact hyperbolic diagram of rank at least three. Complete bounded quotient certificates verify twenty-two named rows of the source's finite table and exclude its other four rows. We state both diagram-only and universal-residue minimum conventions. Auxiliary quotient assertions are corrected explicitly; classifications and earlier constructions retained from the literature are identified throughout.

1 Scope, conventions and acquired proofs

Our source is the sixteen-page revised manuscript [1], whose fixed PDF identity and complete clause ledger accompany this article. Its page 5 defines a spherical hypertope by a finite Coxeter diagram; this allows projective quotients. Section 5, page 10, expressly restricts instead to universal spherical residues. We call the first condition *broad* and the second *locally universal*. Neither convention is silently imposed on the source's hyperbolic table. Isomorphism in the source permits reordering types, including duality. Counts of normal kernels for an ordered presentation are a different invariant and will never be called counts of unlabelled hypertopes.

A hypertope is a thin, residually connected incidence geometry. Its group Aut_I of type-preserving automorphisms is distinguished from its group allowing type permutations. Regularity means that Aut_I acts regularly on chambers. For an involutory marking $G = \langle r_i : i \in I \rangle$ write $G_J = \langle r_j : j \in J \rangle$ and $P_i = G_{I\setminus\{i\}}$. The intersection property is $G_J \cap G_K = G_{J\cap K}$ for all $J, K \subseteq I$.

The complete arguments acquired in this article are supplied as separate editable manuscripts in the reading volume, with their finite certificates and distinct reviews. They are: the residue and marking theorem [15]; the quotient-kernel correction [16]; the spherical argument [17]; the affine kernel and period arguments [18, 19]; finite residue-preserving quotients [20]; the eight-diagram finite core and named groups [21, 22]; the exact Coxeter/LINS interfaces [23]; the ninth-square and exact-order exclusions [24, 25]; and the infinite cubic example [26]. The present proofs make the remaining elementary source joins. Acquiring these complete proofs does not reprove their explicitly stated classical or computational premises.

2 Markings, cosets and the quotient correction

We use the proved residue theorem [15]: the distinguished chamber involutions of a regular hypertope generate its full type-preserving group, satisfy the intersection property, and identify all flag stabilizers with the corresponding marked subgroups. Conversely a C-group whose actual coset incidence system is flag transitive gives a regular hypertope. This is the interface of source

Theorem 2.3; the string C-group/polytope correspondence in source Theorem 2.4 remains the classical Aschbacher–Schulte result.

Proposition 1. *For any family $(P_i)_{i\in I}$ of subgroups of a group G, the typed right cosets P_ig, with incidence given by nonempty intersection, form an incidence system with base chamber $(P_i)_{i\in I}$. Right multiplication acts transitively on flags of each prescribed type of size at most two. Its kernel is $\bigcap_i \mathrm{Core}_G(P_i)$.*

Proof. Cosets of the same subgroup are equal or disjoint, which verifies the same-type incidence axiom. All base cosets contain the identity. Multiplication preserves intersections. For an incident pair choose an element g in its intersection; the pair is then (P_ig, P_jg). This also proves transitivity for singleton flags. An element fixes every coset of type i precisely when it lies in every conjugate of P_i. Intersecting this condition over the types gives the asserted kernel. □

Thus the sentence preceding source Proposition 2.2 must not identify an arbitrary G with the full automorphism group. The proposition's actual action statement is valid. For a regular hypertope the stabilizer/coset theorem supplies the faithful full group, not this general construction alone.

Source Proposition 2.1 and Theorem 3.1 are proved in [16]. In particular, C-group maximal parabolics and all the expected pairwise maximal intersections imply the full intersection property. An epimorphism to a C-group that is injective on all maximal marked parabolics therefore lifts that property. This implication points *upward*; it does not give a reverse homomorphism from a quotient to its cover.

The exact downward criterion, under local injectivity, is

$$N \cap P_i = 1 \quad \text{for all } i, \qquad G/N \text{ is a C-group} \iff N \cap P_iP_j = \{1\} \quad (i \neq j). \tag{1}$$

Here G itself is a C-group and $N \triangleleft G$; the proof lifts two equal coset images into P_i and P_j and applies the maximal-parabolic criterion. Flag transitivity must still be checked. The affine $\widetilde{A}_2$ example in [16] has $G = \mathbb{Z}^2 \rtimes S_3$ and $N = \mathbb{Z}^2$. Each maximal parabolic maps injectively onto S_3, yet the quotient incidence system has one object of each type and one chamber. It is flag transitive but not thin. This is a counterexample to source Theorem 3.2, and (1) is its precise intersection repair.

For any regular hypertope, its marked Coxeter cover $W \twoheadrightarrow G$ and coset reconstruction identify the hypertope with the incidence quotient of the universal complex. Local injectivity follows if its proper residues are universal, since their chamber groups have the same finite orders as the covering parabolics. It does not follow from a finite residue diagram alone. The order-1920 square in Section 6 has local groups of order 60, whereas the covering H_3 parabolics have order 120. It is an explicit broad counterexample to the local-injectivity clause of source Theorem 3.4. The marked quotient and incidence reconstruction survive.

Proposition 2. *A regular hypertope whose diagram has connected components $I_1, \dots, I_t$ is the direct sum of the corresponding regular hypertopes. Its distinguished group is $G_{I_1} \times \cdots \times G_{I_t}$.*

Proof. Marks in different components commute. The intersection property gives $G_{I_a} \cap G_{I\setminus I_a} = 1$. Multiplying the commuting factors therefore gives an injective homomorphism from their direct product onto G. For $i \in I_a$ the maximal parabolic equals the product of $(G_{I_a})_{I_a\setminus\{i\}}$ and all other factors. Its cosets are thus exactly the cosets in the ath factor. Cosets belonging to different components always intersect; within a component their incidence is its factor coset incidence. This proves the direct-sum description. Fixing the base objects outside I_a identifies that factor with a residue of the original hypertope. It is consequently a regular hypertope by the acquired residue theorem. □

The two geometric examples

For source Example 2.5, colour unit cubes of $\mathbb{Z}^3$ by the parity of their lower vertex. The exact dual-cell construction [26] sends the two cube colours to even and odd vertices of the shifted dual grid, source edges to dual squares, and source vertices to dual cubes. Source types $(0, 2, 1, 3)$ become types $(0, 1, 2, 3)$. The direct infinite-grid proof gives flag extension, thinness, residual connectivity and a regular chamber action of $D \rtimes B_3$, where $D = \{v \in \mathbb{Z}^3 : \sum v_i \equiv 0 \pmod 2\}$. Its marks are $\tau_{(1,1,0)} z_1 z_2 s_1, s_1, s_2, z_3$, with z_i sign changes and s_i adjacent coordinate swaps. This is the zero-period object itself; it is not obtained by setting a finite parameter restricted to $t \geq 2$ to zero.

For precision, the negative flag in [17] also proves the nongeometry assertion of source Example 3.3. In the usual D_4 model let $v = (1, 1, 1, 1)/2$, $w = (1, 1, 1, -1)/2$, and $x = e_4$. Their antipodal classes are pairwise incident. A fourth-type object is represented by $y = \pm e_i \pm e_j$. Incidence with $[x]$ requires that one coordinate be 4. Incidence with $[v]$ then requires the two signs of y to agree, whereas incidence with $[w]$ requires them to disagree. Thus this flag has no fourth-type extension. This strengthens the same signed-triangle obstruction from failure of the proposed regular action to failure of flag extension in this rank-four example.

3 The complete spherical classification

Retain Maxwell's Theorem 0.1 [2], as quoted in source Theorem 4.1: a normal subgroup of an irreducible finite Coxeter group is the central antipodal subgroup or the kernel of a graph homomorphism. The primary full proof is not acquired. A graph homomorphism sends simple generators to simple generators or to the identity and covers the target simple generators. In a hypertope quotient the marks remain distinct, nontrivial, and preserve every pair order. Such a graph homomorphism is an isomorphism of the defining Coxeter presentations. The only remaining kernel is the central antipode, when present.

The full argument [17] proves that the antipode avoids every proper parabolic and every product $P_i P_j$. It therefore gives a C-group quotient. The string correspondence pays its geometry for string diagrams. For the other cases, fundamental-weight inner products determine incidence; every quotient flag lifts exactly when its signed triangles have positive sign product. A single explicit triangle excludes every even D_n case. For E_7, E_8, fifteen complete, generator-closed weight orbits, containing 899402 vectors, exclude every bad triangle. Their accepted certificates are acquired, not enumerated again here. The resulting classification is:

Diagram	Range	Universal group order	Additional antipodal object
A_n	$n \geq 1$	$(n+1)!$	no
D_n	$n \geq 4$	$2^{n-1} n!$	no
B_n	$n \geq 3$	$2^n n!$	yes
$I_2(p)$	$p \geq 3$	$2p$	no
E_6		51840	no
E_7		2903040	yes
E_8		696729600	yes
F_4		1152	yes
H_3		120	yes
H_4		14400	yes

An additional antipodal object has half the displayed group order. All proper residues of every object in this table are universal spherical, by the embedded parabolics and exact residue reconstruction. This proves source Theorems 4.2–4.3, subject to the named retained inputs, without assuming local injectivity in order to obtain it.

The E_6 entry corrects source Table 1's $12 \cdot 6!$ to $72 \cdot 6!$. Its exact certificate has a 27-element fundamental-weight orbit with standard D_5 stabilizer of order 1920, giving $27 \cdot 1920 = 51840$.

Conventional extended Schläfli notation in source Table 1 is retained as notation for these universal/projective objects. No additional presentation theorem is inferred from that notation. The source's blanket half-power description of a longest element is not used: in standard A_3, $c = (1234)$ has $c^2 = (13)(24)$, while the longest element is $(14)(23)$. Only actual central antipodes, with the exact powers verified in [17], occur in the proof.

4 Affine kernels and the displayed families

Let W be an irreducible affine Coxeter group in its faithful Euclidean action, and let $N \triangleleft W$ meet every maximal standard parabolic trivially. The complete proof [18] first shows that N is torsion-free: a finite-order isometry fixes a point, which can be moved into the closed alcove and hence has a proper parabolic stabilizer. For $g(x) = Ax + b \in N$, the commutator $gs_ig^{-1}s_i$ is the product of reflections in gM_i and M_i. Nonparallel mirrors give a nontrivial finite-order element, so each wall-normal line is preserved by A. Connectedness of the nonorthogonality graph of the wall normals forces $A = I$ or $-I$. The latter makes g an involution. Thus N consists of translations. This also covers the one-dimensional case directly.

Consequently source *Corollary* 5.1 holds for the locally universal condition explicitly imposed in its Section 5: the hypertope is an incidence quotient by a normal translation subgroup. Maxwell's affine normal-subgroup classification is not required by this proof. The conclusion is necessary, not sufficient: arbitrary translation subgroups can fail (1) or full flag validity. The affine A_2 example above also qualifies the source's page 11 assertion that avoiding individual parabolics completes quotient classification. Euclidean diagrams with projective residues, such as the source's attributed Schulte/Hartley examples, remain outside this locally universal kernel conclusion.

The displayed family classifications are retained with their exact domains. Coxeter [5] and Cacciari [6] give, up to duality,

$$\{4,4\}_{(s,0)},\ \{3,6\}_{(s,0)}\ (s \geq 2), \qquad \{4,4\}_{(s,s)},\ \{3,6\}_{(s,s)}\ (s \geq 1),$$

and $(3,3,3)_{(s,0)}, (3,3,3)_{(s,s)}$ with $s \geq 2$ in both cases. McMullen–Schulte [7, 4] supply the cubic families $\{4, 3^{n-2}, 4\}_{(s^k,0^{n-k})}$ for $s \geq 2$, $k = 1, 2, n$, and the F_4 families $\{3,3,4,3\}_{(s^k,0^{4-k})}$ for $s \geq 2$, $k = 1, 2$. These are attributed classification inputs, not new lattice exhaustions.

For the two nonstring rank-four diagrams the literal adapter [19] proves existence and marked groups of all six displayed families. Put $E = 2\mathbb{Z}^3 + \mathbb{Z}(1,1,1)$; B_3^+ is the signed permutation subgroup with an even number of sign changes. Their groups are $(D/L) \rtimes B_3$ and $(D/L) \rtimes B_3^+$, respectively:

Period and domain	L	$\lvert G_{\widetilde{B}_3}\rvert$	$\lvert G_{\widetilde{A}_3}\rvert$
$(2s,0,0)$, $s \geq 2$	$2s\mathbb{Z}^3$	$192s^3$	$96s^3$
$(s,s,0)$, $s \geq 2$	sD	$48s^3$	$24s^3$
$(2s,2s,2s)$, $s \geq 1$	$2sE$	$768s^3$	$384s^3$

Actual halving geometries, including the smallest allowed periods, supply thinness, residual connectivity and full regularity. The claim that these families exhaust the rank-four cases is retained from Ens [8]. The source's announced all-rank geometric classification in its reference 21 is an announcement, not a proved result acquired here.

5 Compact diagrams and finite existence

The finite, affine and compact hyperbolic simplex diagram classifications are retained from Humphreys [3] and McMullen–Schulte [4]. The exact twenty-one printed rows of source Table 2 are recorded in the clause ledger. Its ten affine rows are $\widetilde{A}_{n-1}$ $(n \geq 3)$, $\widetilde{D}_{n-1}$ $(n \geq 5)$, $\widetilde{C}_{n-1}$

$(n \geq 3)$, $\widetilde{B}_{n-1}$ $(n \geq 4)$, $\widetilde{E}_6, \widetilde{E}_7, \widetilde{E}_8, \widetilde{F}_4, \widetilde{G}_2, \widetilde{A}_1$. The last is rank two with an infinity bond. The eleven hyperbolic rows comprise the strings $[p, q]$ with $1/p + 1/q < 1/2$, and triangles (k, l, m) with $1/k+1/l+1/m < 1$; the rank-four squares $(3, 3, 3, p)$, $p = 4, 5$, $(3, 4, 3, 4)$, $(3, 5, 3, p)$, $p = 4, 5$; strings $[p, 3, 5]$, $p = 4, 5$, and $[3, 5, 3]$; the fork $[5, 3_3]$; the rank-five pentagon $(3, 3, 3, 3, 4)$; strings $[p, 3, 3, 5]$, $p = 3, 4, 5$; and the fork $[5, 3, 3_3]$. All unspecified finite labels in this description are at least three. Repeated parameter values give nine rank-four and five rank-five diagrams.

Theorem 3 (Acquired finite-existence theorem)**.** *If an infinite Coxeter system has finite rank at least three and every maximal standard parabolic is finite, it has finite regular hypertope quotients of unbounded chamber number with unchanged diagram and universal spherical proper residues.*

The proof [20] avoids every nonidentity element in the finite union $\bigcup_{i,j,k} P_i P_j P_k$. Two factors ensure the full intersection property; three factors ensure that pairwise incident quotient objects represented within one parabolic were incident before quotienting. Every flag then lifts and each full residue incidence is preserved. In the faithful geometric representation all coefficients lie in a fixed ring of algebraic integers. A rational prime avoiding finitely many nonzero norms separates this finite set by reduction. Separating arbitrarily many further elements gives unbounded chamber numbers. This proves existence for all retained compact diagrams, including nonstrings, but supplies no small-order bound. Rank two is expressly excluded: a finite quotient of the infinite dihedral group changes its infinity bond.

The source's earlier construction descriptions remain individually attributed. Corn–Singerman [9] provide the regular-hypermap triangle-rotation presentation; Monson–Schulte [10] the modular $\{3, 5, 3\}$ examples; Breda–Jones–Schulte [11] the $D_s^6 \rtimes [3, 5]$, $s \geq 2$, construction; Hartley–Leemans [12] the $J_1 \times J_1$ and $J_1 \times J_1 \times L_2(19)$ examples. The star-polytope construction of McMullen [13], also described by Coxeter [14], has the stated rank-four 120 facets and vertex figures and group H_4 of order 14400. Its deep-hole relations are retained as part of that specific construction. The rank-five deep-hole object cited on source page 13 is an infinite 5-apeirotope; that page does not explicitly call it finite. Theorem 3 proves finite existence independently and does not reproduce these particular groups or historical constructions.

6 The complete finite-table disposition

For a square (a, b, c, d) the four cycle edges have those labels and the diagonals label two. Strings $[a, b, c]$ have three consecutive bonds, and $[5, 3_3]$ is the rank-four fork with bonds $5, 3, 3$. This fixes the ordered Coxeter presentations up to the source's permitted diagram relabelling. Every regular hypertope with such a diagram gives a normal kernel in this presentation of index equal to its number of chambers. No assumption of universal local residues enters this necessary enumeration interface.

The eight-type core [21] completely checks all 120 normal kernels through its eight respective bounds and obtains 59 positive marked kernels. Its test checks all marked subgroup intersections and all sets of objects incident with each base flag. Induction on flag size proves that this tests *all* actual flags, not only those already known to lie in chambers. It therefore pays flag extension, thinness, residual connectivity, and the full type-preserving group. The separate structural certificates in [22] prove exactly the following twenty-two source group rows. Here split actions are the actual certified actions; notation alone is not a uniqueness or isomorphism proof.

Diagram	Certified source group	Order
$(3, 3, 3, 4)$	$2^4 : A_5 : 2$	1920
$(3, 3, 3, 4)$	$(2^3 : L_2(7)) \times C_2$	2688
$(3, 3, 3, 4)$	$2^5 : A_5 : 2$	3840
$(3, 3, 3, 4)$	$(4 \cdot (2^4 : A_5)) : 2$	7680

Diagram	Certified source group	Order
$(3,3,3,4)$	$2^6 : A_5 : 2$	7680
$(3,3,3,5)$	$(A_5 \times A_5) : 2$	7200
$(3,4,3,4)$	$2^{1+4} : S_3 : S_3$	1152
$(3,4,3,4)$	$S_6 \times C_2$	1440
$(3,4,3,4)$	$S_6 \times S_3$	4320
$(3,4,3,4)$	$(2^3 : L_2(7)) \times C_2^2$	5376
$(3,4,3,4)$	$L_2(7) \times S_3^2$	6048
$(3,4,3,4)$	$2^8 : S_3 : S_3$	9216
$[3,5,3]$	$L_2(16) : 2$	8160
$[4,3,5]$	$2^5 : A_5 : 2$	3840
$[4,3,5]$	$(2^4 : A_5 : 2) \times C_2$	3840
$[4,3,5]$	$(2^5 : A_5 : 2) \times C_2$	7680
$[4,3,5]$	$2^5 : A_5 : 2 : 2$	7680
$[5,3,5]$	$(A_5 \times A_5) : 2$	7200
$[5,3,5]$	$L_2(16) : 2$	8160
$[5,3_3]$	$(2^5 : A_5) \times C_2$	3840
$[5,3_3]$	$2^6 : A_5$	3840
$[5,3_3]$	$(2^6 : A_5) \times C_2$	7680

The displayed nonsplit dot has an actual central/derived obstruction to a complement. Split and direct factors have literal certificates, including the affine $2^3 : L_2(7)$ model, extraspecial 2^{1+4}, and the outer involution in $L_2(16) : 2$. Every row in this positive table has universal spherical local groups.

Four printed rows are excluded even under the broad convention:

Diagram	Printed order	Complete exclusion
$(3,4,3,4)$	336	all three exact-order kernels fail IP
$(3,5,3,4)$	2160	the exact-order kernel fails IP
$(3,5,3,5)$	3840	all eleven exact-order kernels fail IP
$(3,5,3,5)$	9720	the exact-order kernel fails IP

The first two follow from [21]. The complete ninth-square search through 3840 [24] returns 25 kernels and 23 markings after the necessary rank-two divisibility filter. Twenty-one fail the intersection property. The order-960 candidate satisfies it but has a nonextendible three-type flag. The sole positive has order 1920 and all four local orders 60. Its explicit permutation tuple and full incidence proof are supplied; its global abstract structural name is not asserted.

For order 9720, the exact-divisor method [25] preserves every normal overgroup of a sought kernel: its index must divide 9720. The completed search has the six indices $1, 2, 60, 120, 4860, 9720$, each once. Its sole exact-order quotient fails twelve marked intersections. An independent integer closure exhibits the same nonidentity central involution in all four maximal parabolics. This excludes that row without repeating the timed-out unrestricted search through 9720, and makes no completeness claim at unrelated indices.

The complete bounded minima, including all source-equivalent markings, are:

Diagram	Complete bound	Broad minimum	Universal minimum
$(3,3,3,4)$	7680	168	1920
$(3,3,3,5)$	7200	2520	7200
$(3,4,3,4)$	9216	168	1152
$(3,5,3,4)$	2160	1080	> 2160
$(3,5,3,5)$	3840	1920	> 3840
$[3,5,3]$	8160	660	8160
$[4,3,5]$	7680	1920	3840
$[5,3,5]$	8160	3420	7200
$[5,3_3]$	7680	960	3840

The broad minima are all attained. An inequality in the last column means only exclusion through that bound; no attained universal minimum beyond it is claimed. Source Table 3's assertion that each first row is smallest is therefore replaced by this explicit two-convention table. For example the product-size obstruction $48^2/6 = 384 > 336$ applies only to universal B_3 residues; the full three-kernel test is what excludes 336 without that hypothesis. Kernel inventories retain all multiplicities and are compared by exact ordered-generator isomorphisms, not compact permutation byte equality or abstract group names.

These finite exhaustions retain the correctness and completeness contract of GAP/LINS, its default normal-subgroup traversal, and its fixed tables of characteristically simple groups, multiplier primes and socle overgroups. The installed maximum index 10^7 exceeds every bound used. The acquired Coxeter seed and module-interface proofs [23] and divisor proof state the precise retained interfaces. Literal closures and coset readers independently verify returned markings, but do not constitute a formal verification of the normal-subgroup algorithm or its classification tables.

7 Source disposition, premises and attribution

The clause ledger contains 109 entries: the original 83 stable source clauses, the 21 explicit Table 2 rows, and five scope clauses. It routes all numbered statements, both examples, all ten Table 1 rows, all affine families, all twenty-six Table 3 rows and nine minimum assertions. Corrections are local to their stated assertions: Theorem 3.2; local injectivity in the broad reading of Theorem 3.4; E_6's order; the overbroad Coxeter-element half-power sentence; sufficiency of affine parabolic avoidance; and the four finite-table rows and consequent minimum claims. The source's rank-five small search remains an inconclusive historical search, not a nonexistence theorem. No formal named conjecture is disposed of here.

The indispensable retained inputs are the classical Coxeter complex and parabolic/chamber facts; faithful geometric representations and affine alcove stabilizers; finite/affine/compact diagram classifications; Maxwell's finite normal-subgroup alternative; the string correspondence; the named toroidal classifications; the cited earlier special constructions; and the exact finite software/classification-table contracts. Their roles and source citations are explicit in the ledger. In particular, neither a new all-rank affine lattice classification nor independent reconstructions of every historical example are claimed. The source's abstract must be read with its Section 5 review-and-announcement boundary.

The assembly checker verifies the exact acquired manuscripts, certificates, reviews and their declared dependencies, the source-table joins, and the native document. Small independent controls exercise the new direct-sum, coset-kernel and D_4 joins and reject damaged ledgers. It does not rerun an acquired group census or exceptional weight-orbit enumeration. Mathematical validity of the new joins requires the separate reading of this proof; hash identity is only an acquisition check.

The classifications and source questions belong to the cited authors. Component authorship and distinct review roles are preserved in their records. The present author previously

contributed the eight-type finite core, named-group certificates and Coxeter interfaces, and independently reviewed the separate divisor, ninth-square and cubic applications. The present synthesis and elementary joins require their own distinct review. No claim of historical novelty, new formalization, new question credit or measured compression follows from this assembly.

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[23] Human Compression project, complete reviewed Coxeter seed and LINS composition interfaces, L033 method component.

[24] Human Compression project, complete square $(3, 5, 3, 5)$ quotients through order 3840, L033 ninth-square component.

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